SHIV NADAR UNIVERSITY

Programme: **B. Tech** Discipline: **Computer Science and Engineering**
Exam: **Mid Semester** Year: **2019-2020**
Course Code: **CSD301** Course Title: **Software Engineering**
Date: **September 24, 2019** Time: **02:30 PM – 04:00 PM** Max. Marks: **40**

Ques. 1 The later a problem is found in the software development process, the harder it is to fix. Explain why, in one sentence. **(1 Mark)**

Ques. 2 In COCOMO Model, if project size is typically 2-50 KLOC, then which mode is to be selected? (Organic, Semi-Detached, Embedded) **(1 Mark)**

Ques. 3 What is the difference between Product and Process in software terminology? **(1 Mark)**

Ques. 4 Among various properties of software, one is "Software does not wear out". Justify this statement by comparing software with the "bath tub curve" for hardware components. **(2 Marks)**

Ques. 5 What are the components of a Use Case Diagram? Explain their usage with the help of an example. **(2.5 Marks)**

Ques. 6 Match each lifecycle model in **'Column A'** with its definition in **'Column B'**. **(2.5 Marks)**

Column A	Column B
(a) code-and-fix	**(i)** assess risks at each step; do most critical action first
(b) evolutionary prototyping	**(ii)** build an initial small requirement spec, code it, then "evolve" the spec and code as needed
(c) spiral	**(iii)** build initial requirement specs for several releases, then design-and-code each in sequence
(d) staged delivery	**(iv)** standard phases (requirements, design, code, test) in order
(e) waterfall	**(v)** write some code, debug it, repeat (i.e. ad-hoc)

Ques. 7 Describe the spiral model of software development. **(5 Marks)**

Ques. 8 A project size of 200 KLOC is to be developed. Software development team has average experience on similar type of projects. The project schedule is not very tight. Calculate the effort, development time, average staff size and productivity of the project. **(5 Marks)**

Ques. 9 Draw **level 0** (Context level) and **level 1 DFD** for the following students' academic record management software. **(5 Marks)**

- A set of courses are created. Each course consists of a unique course number, number of credits, and the syllabus.
- Students are admitted to courses. Each students' details include his roll number, address, semester number and the courses registered for the semester.
- The marks of student for various units he created are keyed in.
- Once the marks are keyed in, the semester weighted average (SWA) is calculated.

- The recent marks of the student are added to his previous marks and a weighted average based on the credit points for various units is calculated.
- The marks for the current semester are formatted and printed.
- The SWA appears on the report.
- A check must be made to determine if a student should be placed on the Dean's list. This is determined based on whether a student scores an SWA of 85 or higher.
- If the SWA is lower than 50, the student is placed on a conditional standing.

Ques. 10 A target product has 7 inputs (simple), 11-average inputs and 8 complex inputs. There are average outputs-5, simple inquiries 18 master files, and 12 complex interfaces. **(5 Marks)**

(i) Determine UFP
(ii) Determine Function Point (FP) if total Degree of Influence is 49.

*Refer Below Table for Refinement of Function Point Entities

Type	Simple	Average	Complex
Input (I)	3	4	6
Output (O)	4	5	7
Inquiry (E)	3	4	6
Number of Files (F)	7	10	15
Number of Interfaces	5	7	10

Ques. 11 Refer the following table. Draw an activity chart showing the project schedule. Find the critical path. **(10 Marks)**

Task	**T1**	**T2**	**T3**	**T4**	**T5**	**T6**	**T7**	**T8**	**T9**	**T10**	**T11**	**T12**
Duration (Days)	8	15	15	10	10	5	20	25	15	15	7	10
Dependencies			T1		T2, T4	T1, T2	T1	T4	T3,T6	T5, T7	T9	T11

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